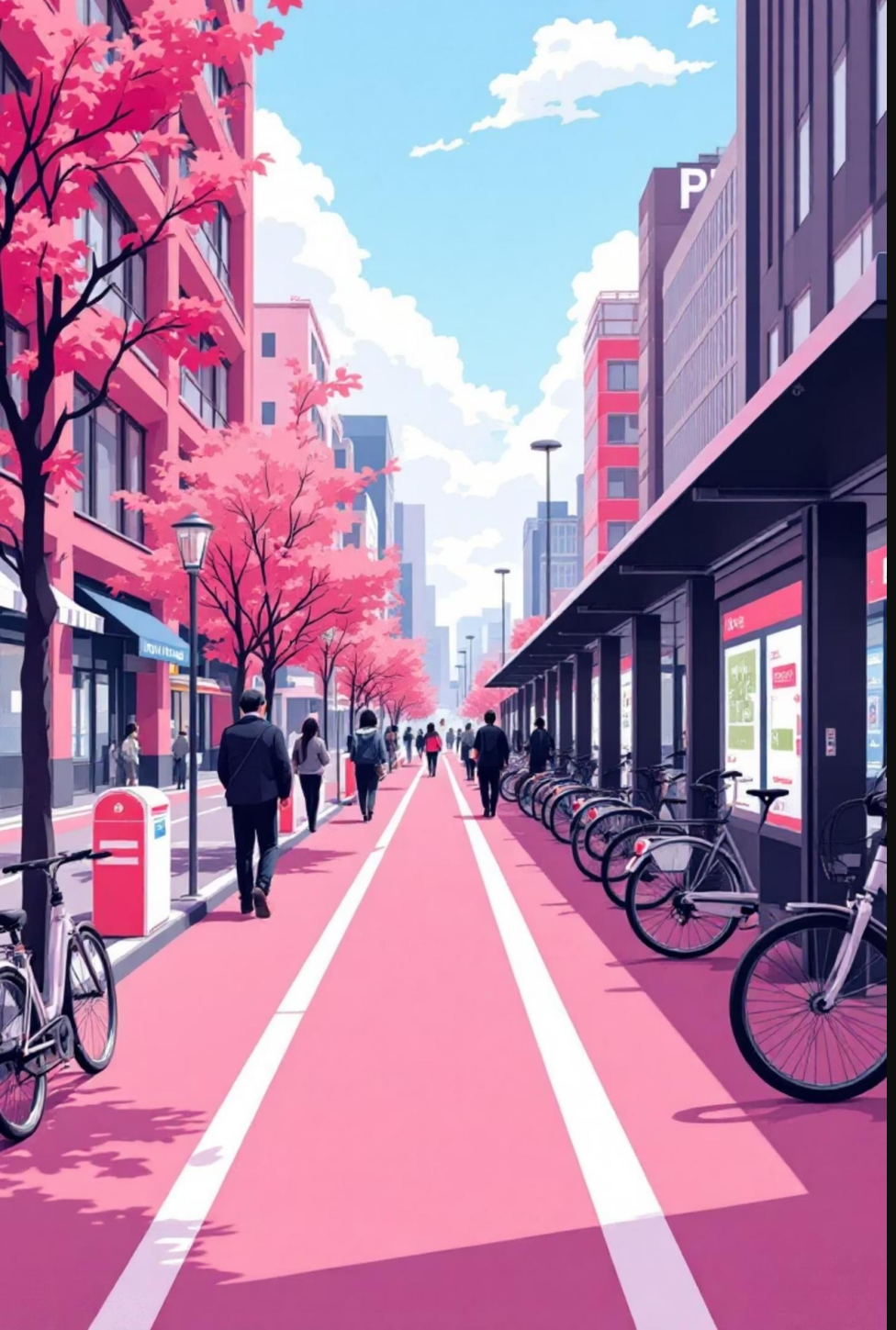


Hourly Bike-Sharing Demand Prediction

Introduction to Bike Sharing

What is Bike Sharing?

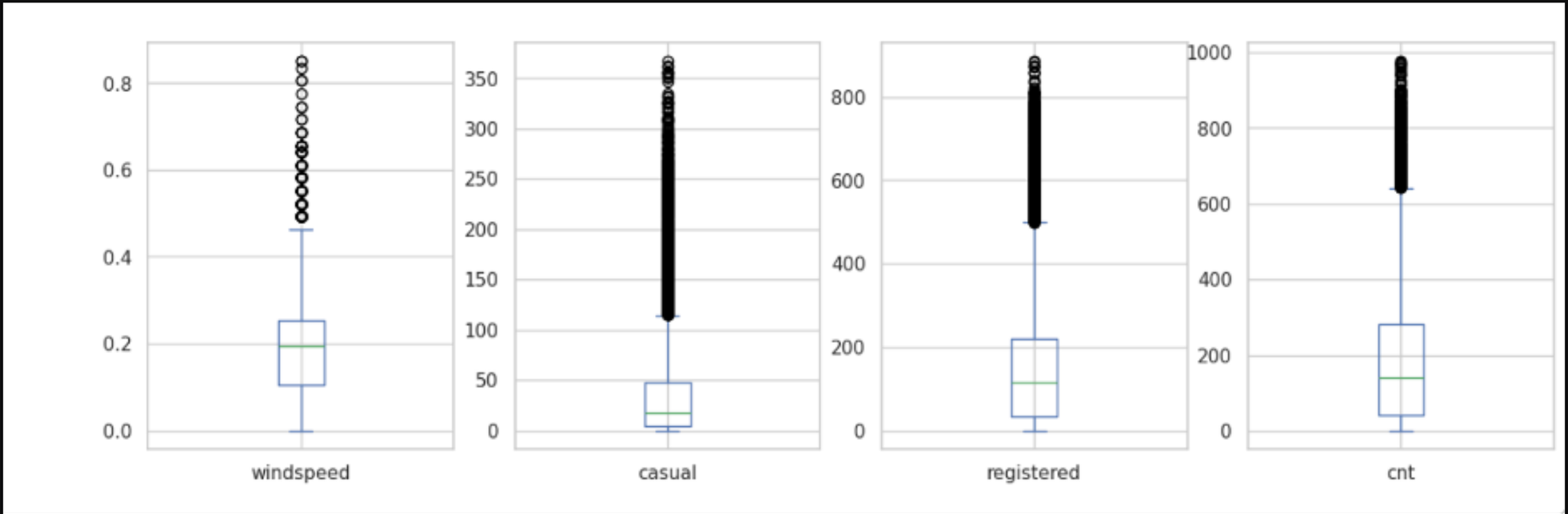
A system where individuals rent bikes for short durations via an app or card, returning them to a station or designated location.

Our Objective

Develop a machine learning model to accurately predict hourly bike-sharing demand using variables like weather, seasonality, and time.

Why it Matters

Accurate prediction is crucial for operational planning, bike redistribution, and efficient resource allocation in bike-sharing systems.




```
Total outliers: 3244
Outliers per column:
 instant          0
season            0
yr                0
mnth              0
hr                0
holiday         500
weekday           0
workingday        0
weathersit        3
temp              0
atemp             0
hum              22
windspeed       342
casual         1192
registered      680
cnt             505
dtype: int64
```

Outliers were identified using the IQR.

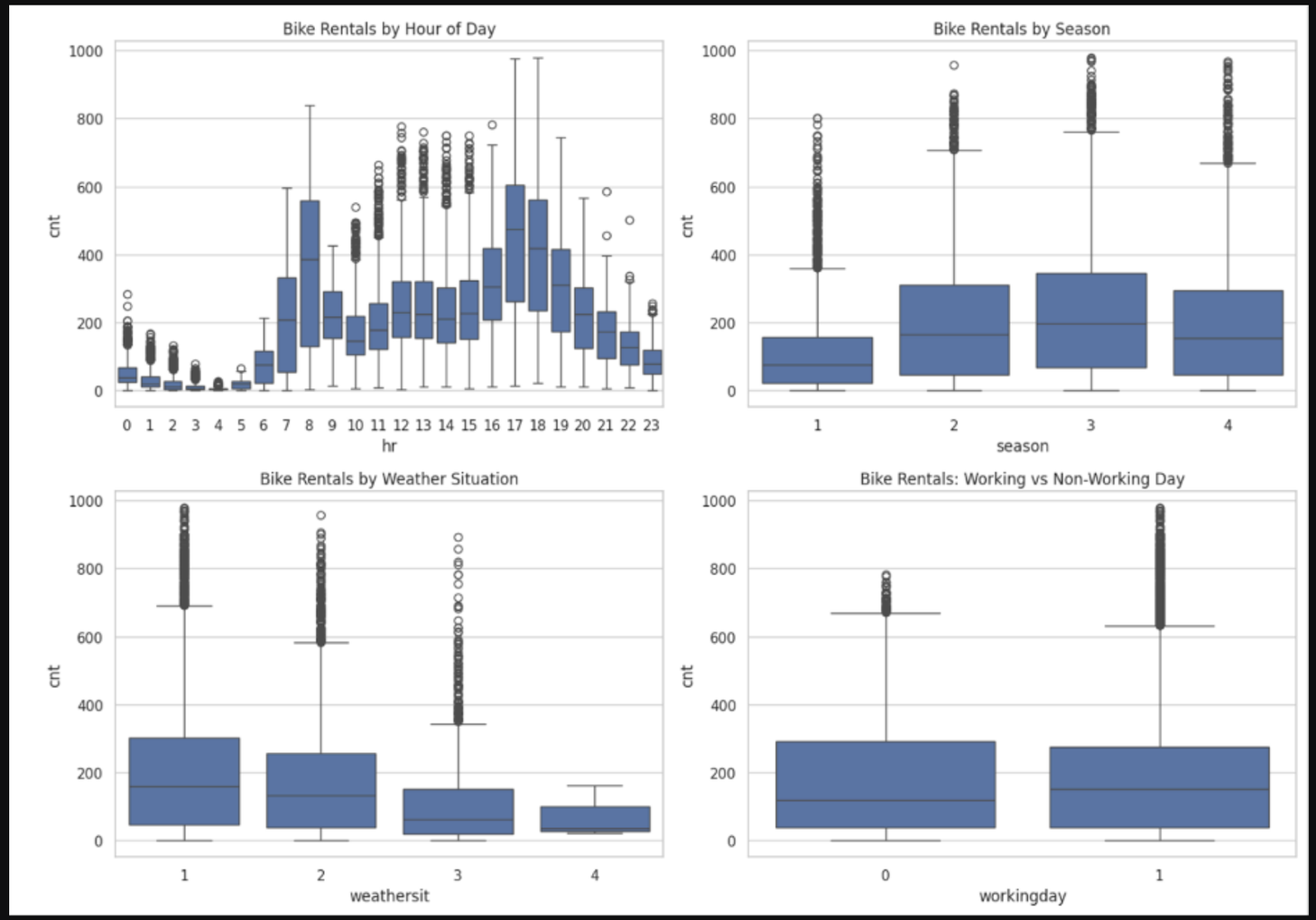
Holiday had around 500 outliers and this variable is binary (0 or 1).
The detected outliers reflect class imbalance rather than true anomalies, since holidays are naturally less frequent than non-holidays.

weathersit (3 outliers): The weather situation has limited categorical values (1–4). The few outliers we have pertain to rare severe weather conditions, which are valid real-world cases.

hum (22 outliers) and windspeed (342 outliers): This represents our extreme humidity and wind conditions.

Demand-Related Variables: casual (1,192 outliers), registered (680 outliers), cnt (505 outliers): These outliers represent the very high or very low bike rental counts. They often occurring during peak commuting hours, favorable weather, special events or seasonal surges. These values are indicating data errors but true demand variability.

Modeling Decision: We did not remove outliers from our dataset. The outliers represented real-world behavior and removing them would affect peak-demand patterns. As a result, we kept the outliers to preserve the realistic demand patterns like rush hours, events, nice weather.



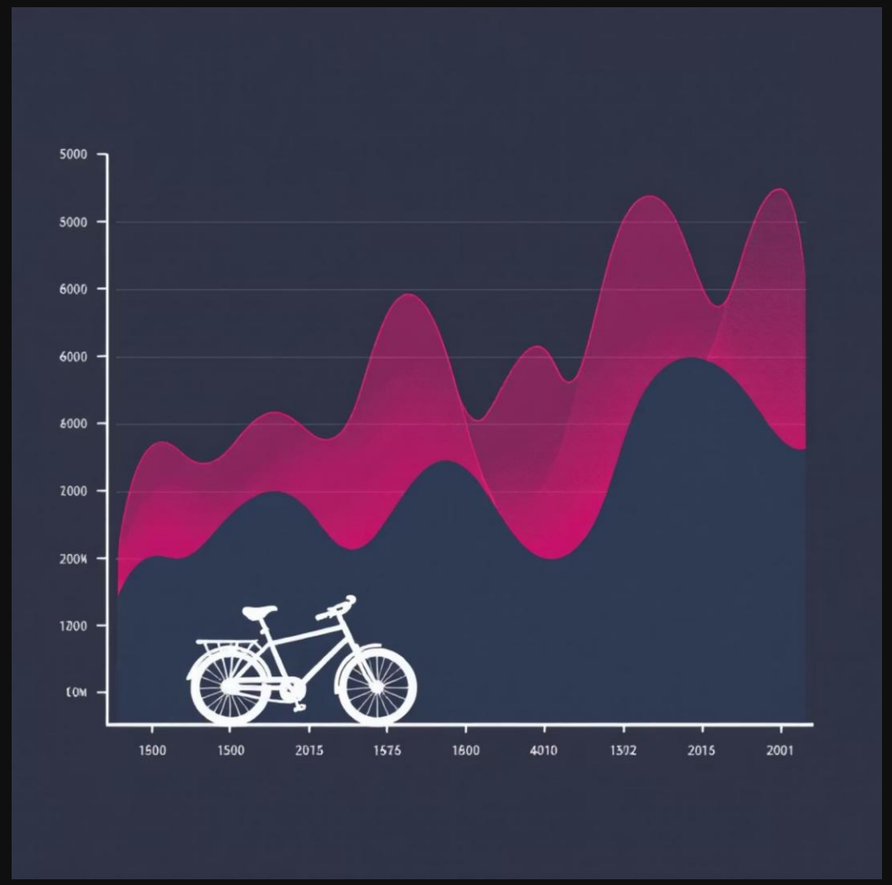
Bike Rentals by Hour of Day: The hourly bike rentals show a bimodal pattern. There are strong peaks during the morning around 7-9 AM and the evening period around 4-7 PM which are commuting hours. Demand is lowest during the late night and early morning hours. This suggests that bike usage is influenced by commuting behaviors. Therefore, our time features are highly predictive of demand.

Bike Rentals by Season: The bike rental demand varies significantly across seasons. There is higher median rentals in warmer seasons like the Spring and Summer. There is lower demand during colder seasons. This indicates that seasonality is playing a major role in bike usage because of the impacts of things like temperature and daylight.

Bike Rentals by Weather Situation: Rental counts declined as the weather conditions worsened. Clear or mildly cloudy weather shows the highest rental demand. Meanwhile, bad weather conditions resulted in lower rentals. This implies that weather has a strong negative impact on bike usage.

Bike Rentals: Working vs Non-Working Days: Working days have higher and more consistent rental demand. Non-working days have more variability with some occasional spikes. This reinforces the idea we discussed earlier on bike usage being influenced by commuting behaviors.

User Distribution & Demand Patterns



Registered users drive most demand, showing a wider range of hourly rentals. Casual users rent less frequently and are more sensitive to conditions, leading to a right-skewed distribution.

Hourly Rentals

Bimodal pattern with peaks during morning (7-9 AM) and evening (4-7 PM) commuting hours. Lowest demand late night/early morning.

Key Factors Influencing Demand

Seasonality

Higher demand in warmer seasons (Spring, Summer), lower in colder seasons due to temperature and daylight impacts.

Weather Conditions

Clear or mildly cloudy weather shows highest demand. Bad weather significantly reduces rentals.

Working vs. Non-Working Days

Working days have higher, more consistent demand, reinforcing commuting influence.

Correlation Analysis

The correlation analysis shows that total rentals (cnt) are strongly positively correlated with registered (0.97) and casual users (0.69). This is expected since they form the total demand together.

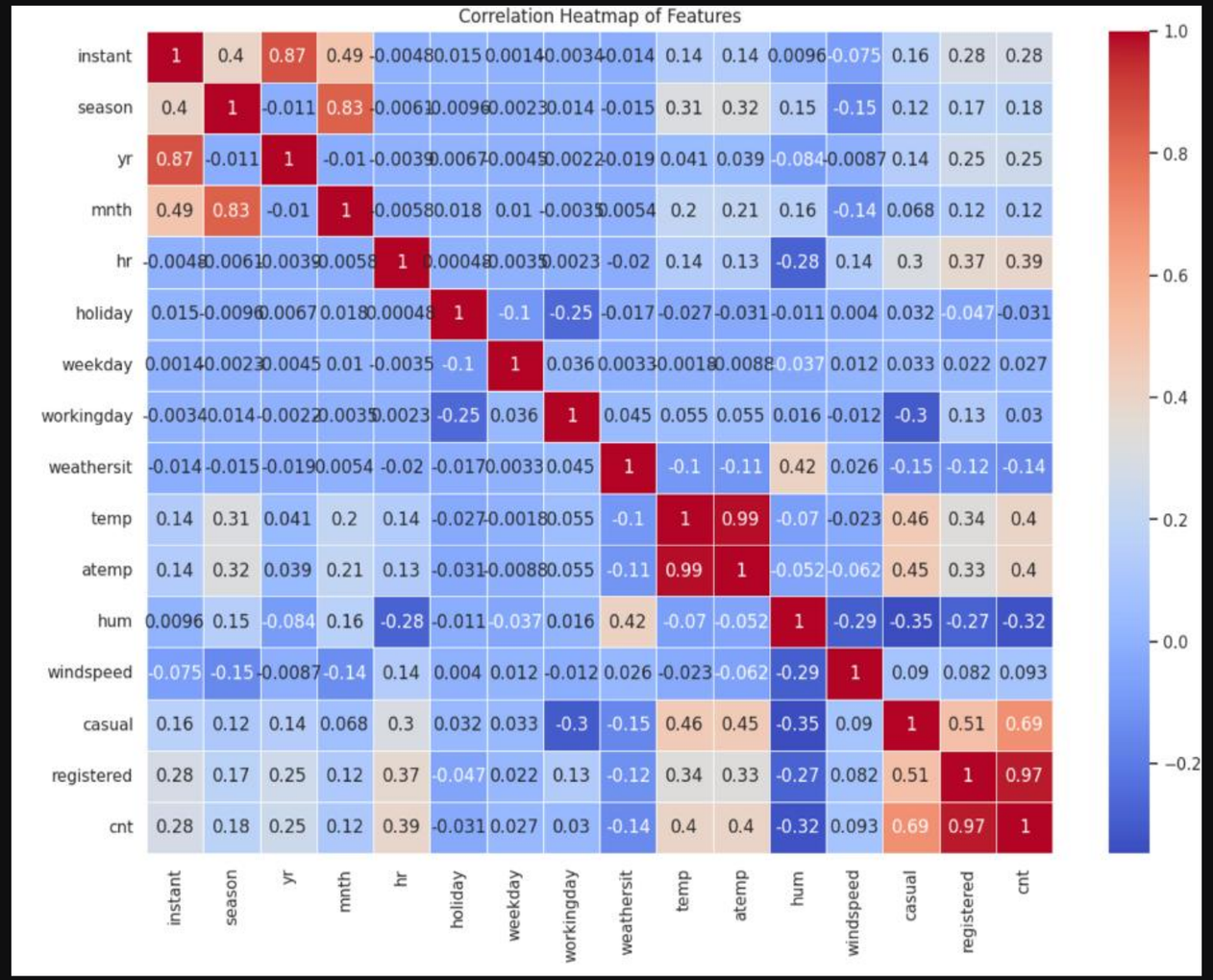
Temperature variables (temp and atemp) are almost perfectly correlated (0.99), indicating strong multicollinearity and potential redundancy for our modeling.

Time-related variables show meaningful relationships, with hour of day (hr) moderately correlated with cnt (0.39) and season and month highly correlated (0.83).

Weather effects are evident, as humidity has a negative correlation with cnt (-0.32) and weather situation also shows a negative relationship (-0.14), suggesting reduced demand under unfavorable conditions.

Variables such as windspeed (-0.003) and weekday/workingday exhibit weak correlations with total rentals.

Overall, the heatmap implies that time and temperature are the dominant drivers of bike-sharing demand, while adverse weather conditions decrease usage.



Models Evaluated & Metrics

1

Decision Tree Regressor

A non-parametric supervised learning method used for both classification and regression.

2

Random Forest Regressor

An ensemble learning method that operates by constructing multiple decision trees.

3

Support Vector Regression (SVR)

Utilizes an RBF Kernel to model non-linear relationships in data.

4

Stochastic Gradient Descent (SGD) Regressor

An iterative optimization algorithm for minimizing an objective function.

Mean Squared Error (MSE)

Penalizes large errors, measuring average squared difference.

Mean Absolute Error (MAE)

Provides intuitive interpretation of average absolute prediction error.

R^2 Score

Indicates proportion of variance in bike demand explained by the model.

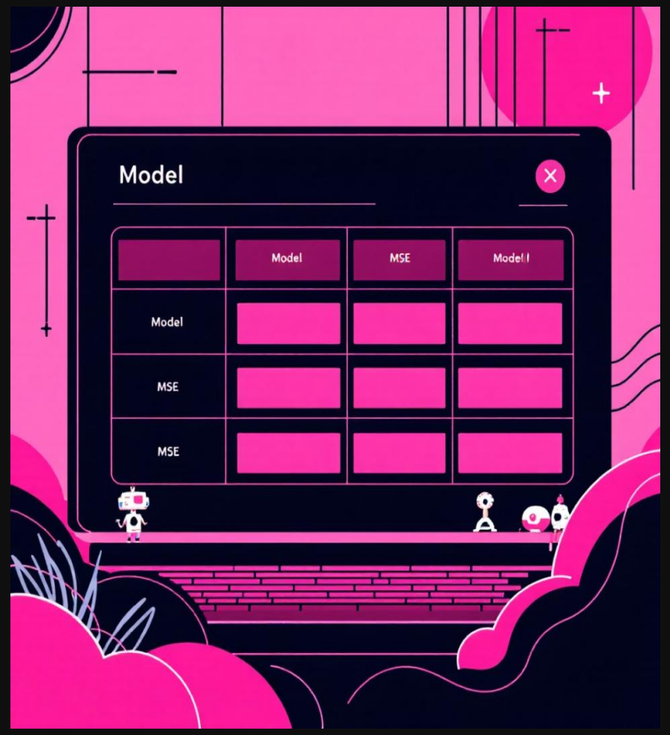
	Model	Val MSE	Val MAE	Val R^2	Test MSE	Test MAE	Test R^2
0	Decision Tree	6405.505408	53.882043	0.803004	6352.227932	53.584863	0.799395
1	Random Forest	2557.967390	31.984364	0.921332	2524.563174	31.280141	0.920274
2	SVR (RBF Kernel)	2094.352830	28.661620	0.935590	2169.112221	29.044629	0.931499
3	SGD Regressor	9844.319855	72.993041	0.697247	10089.143616	74.001671	0.681383

Performance Results

Validation and Test Performance

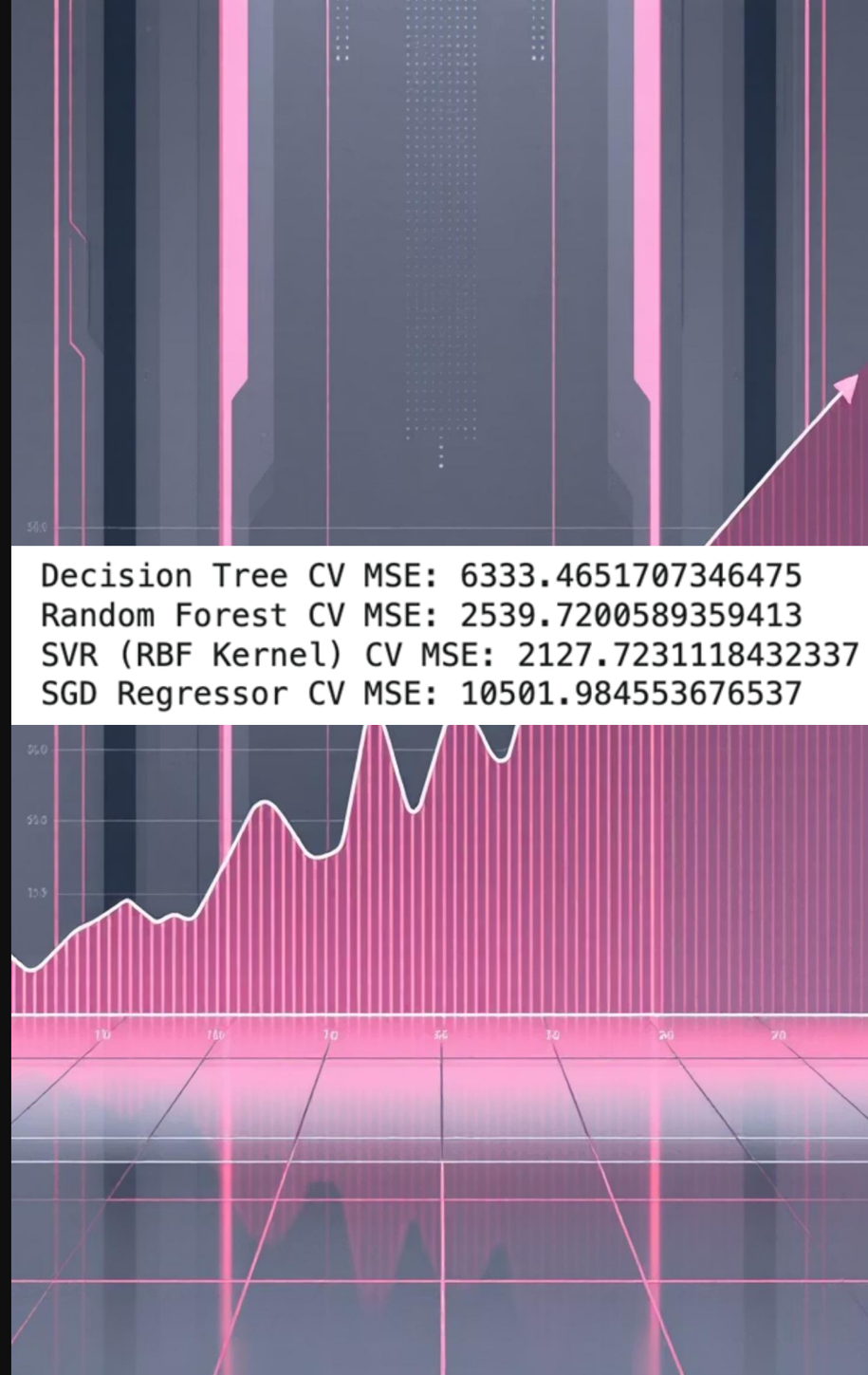
SVR (RBF kernel) shows the best performance across validation and test sets.

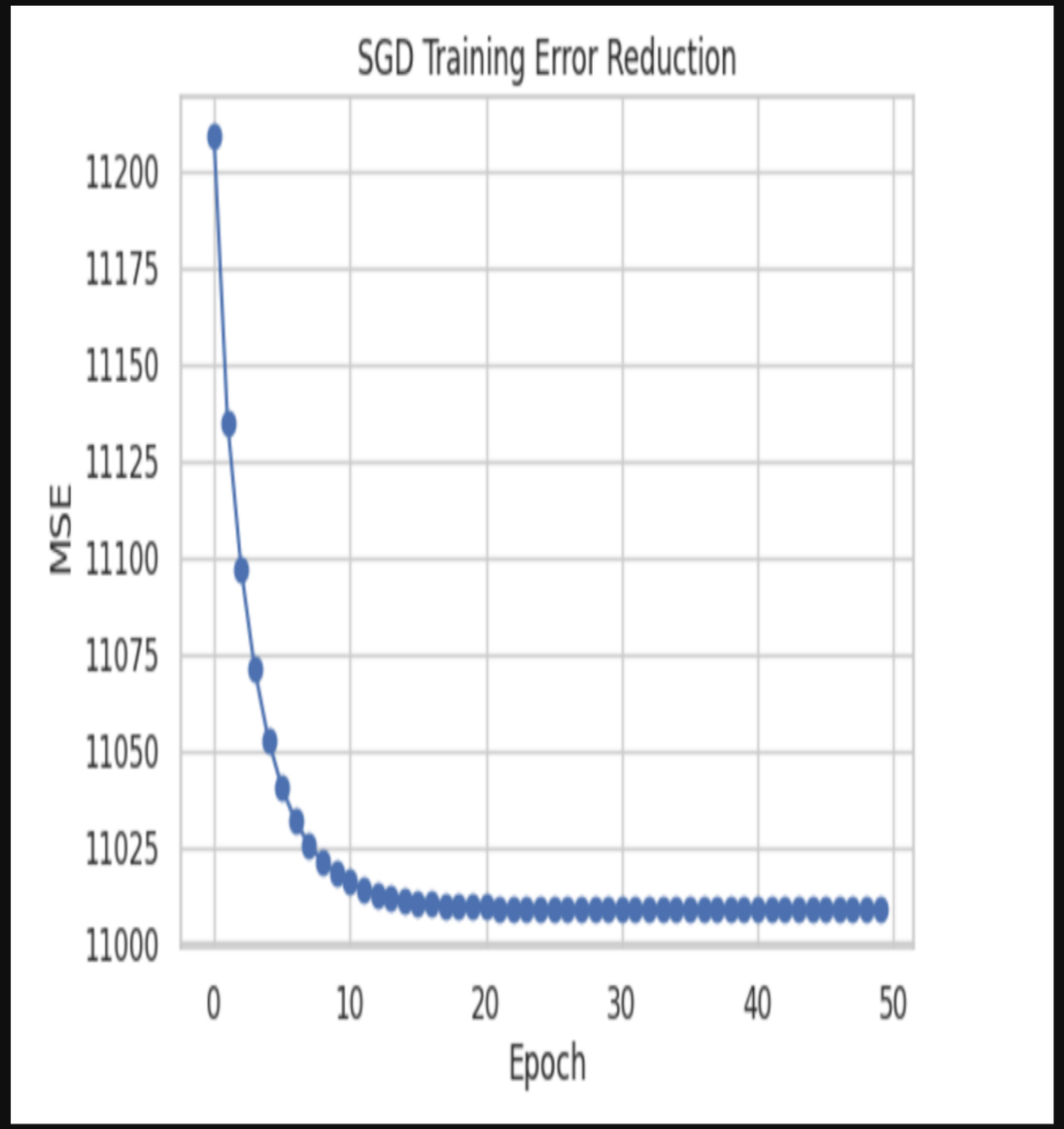
Cross-Validation (K-Fold)



SVR (RBF kernel) performs best with the lowest and most stable MSE across folds. Random Forest is second.

SGD Regressor optimization confirmed by decreasing training error across epochs.





Stochastic Gradient Descent (SGD) was used to optimize the regression model by iteratively updating model parameters to minimize the Mean Squared Error (MSE) between predicted and actual bike rental counts. The decreasing training error across epochs confirms effective optimization and convergence.

Hyperparameter Tuning

Random Forest

Achieved lowest Test MSE (2412.44) with optimal parameters: max_depth=20, min_samples_split=2, n_estimators=200.

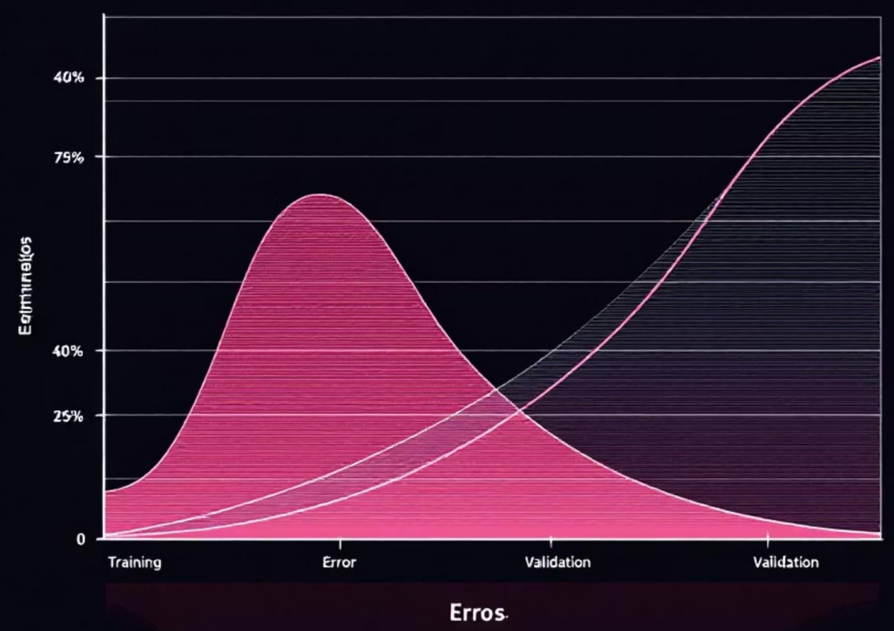
Decision Tree

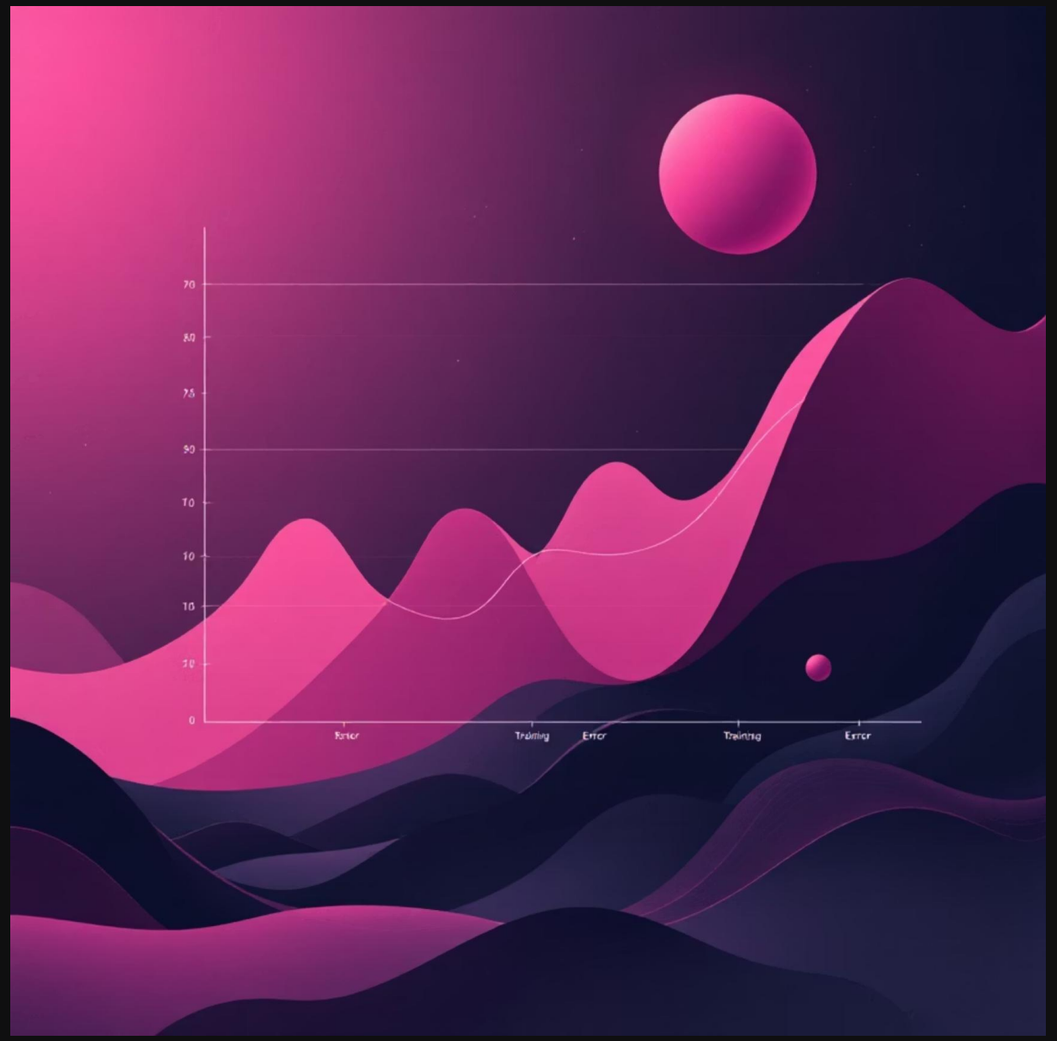
Improved but higher Test MSE (4022.49) with max_depth=20, min_samples_split=10.

SGD Regressor

Remained weakest performer with Test MSE of 10122.98, even after tuning.

Model Learning Curves



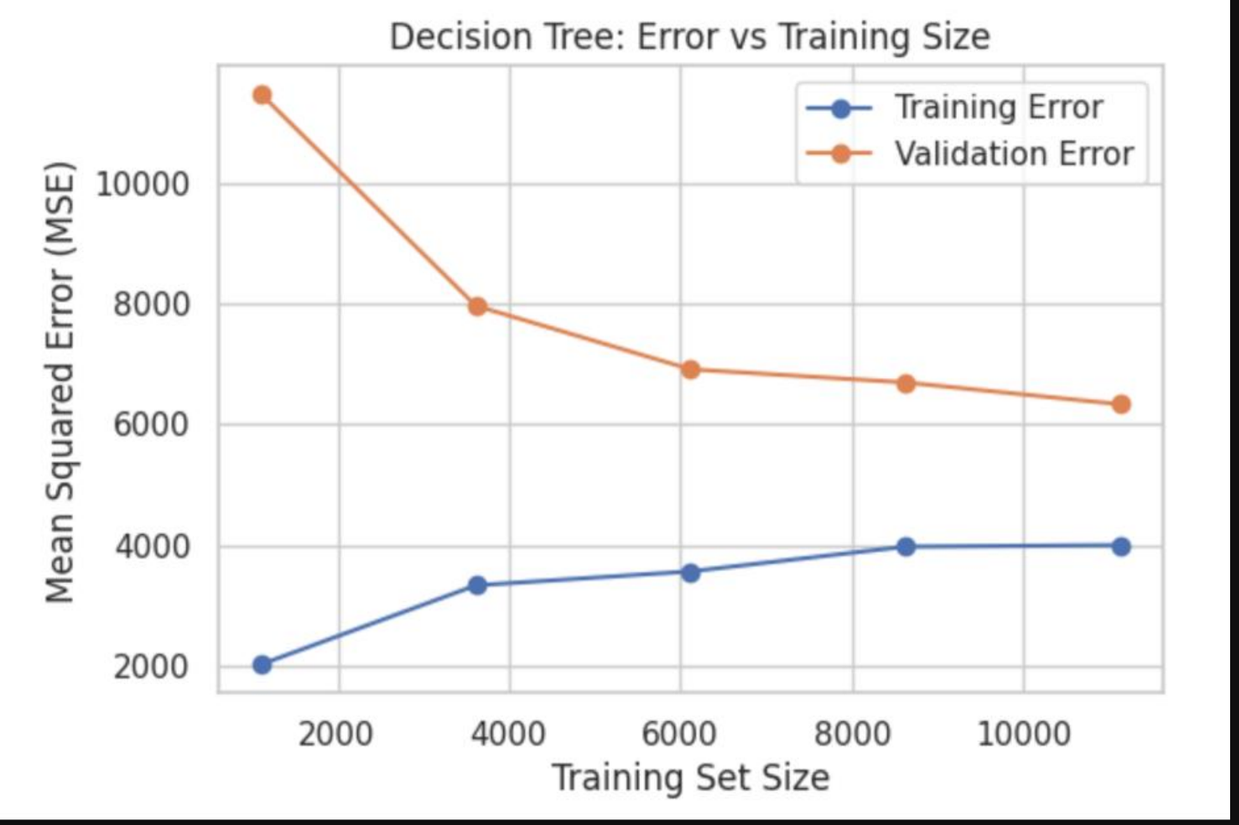


Decision Tree

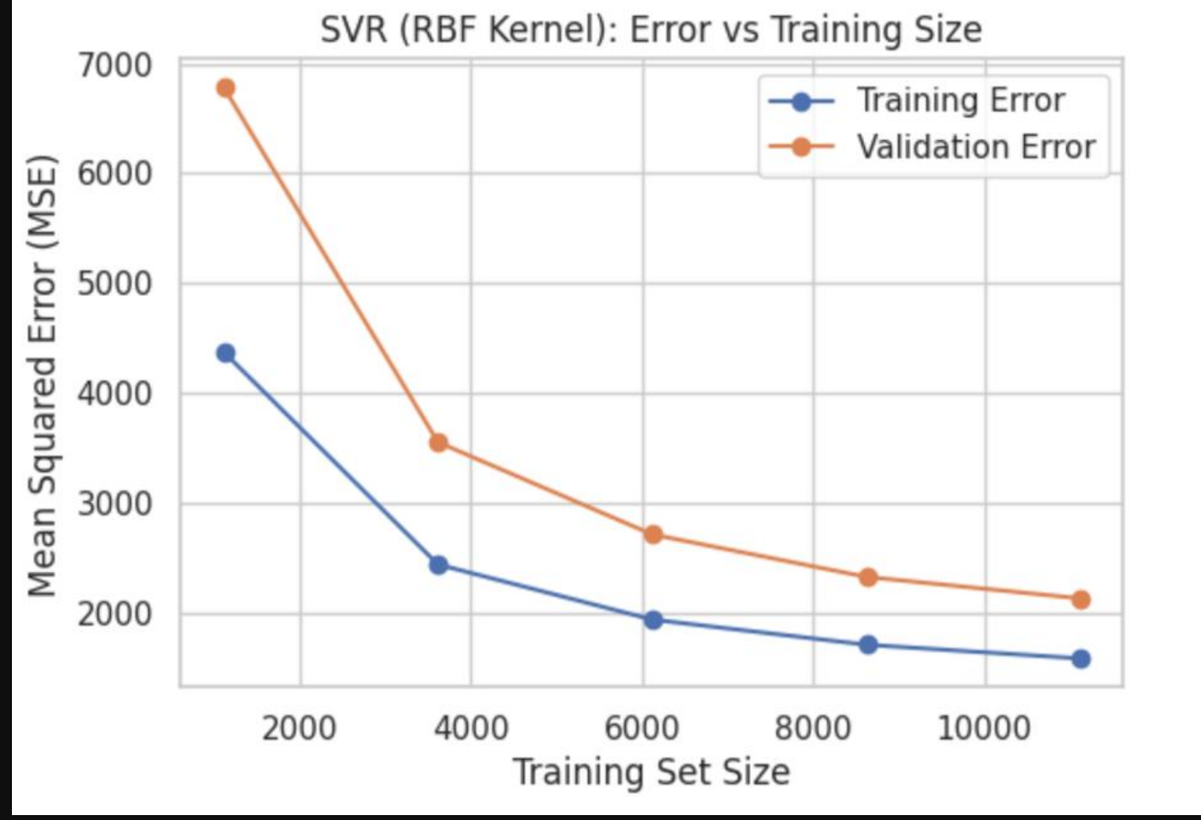
Low training error on small datasets, increasing to 4,000 MSE. Validation error high (>11,000 MSE) decreasing to 6,300–6,500 MSE. Indicates overfitting and high variance.

SVR (RBF)

Steady decrease in training error (4,300 to 1,600 MSE). Sharp drop in validation error (6,800 to 2,100 MSE). Small gap suggests low variance and good bias-variance balance.

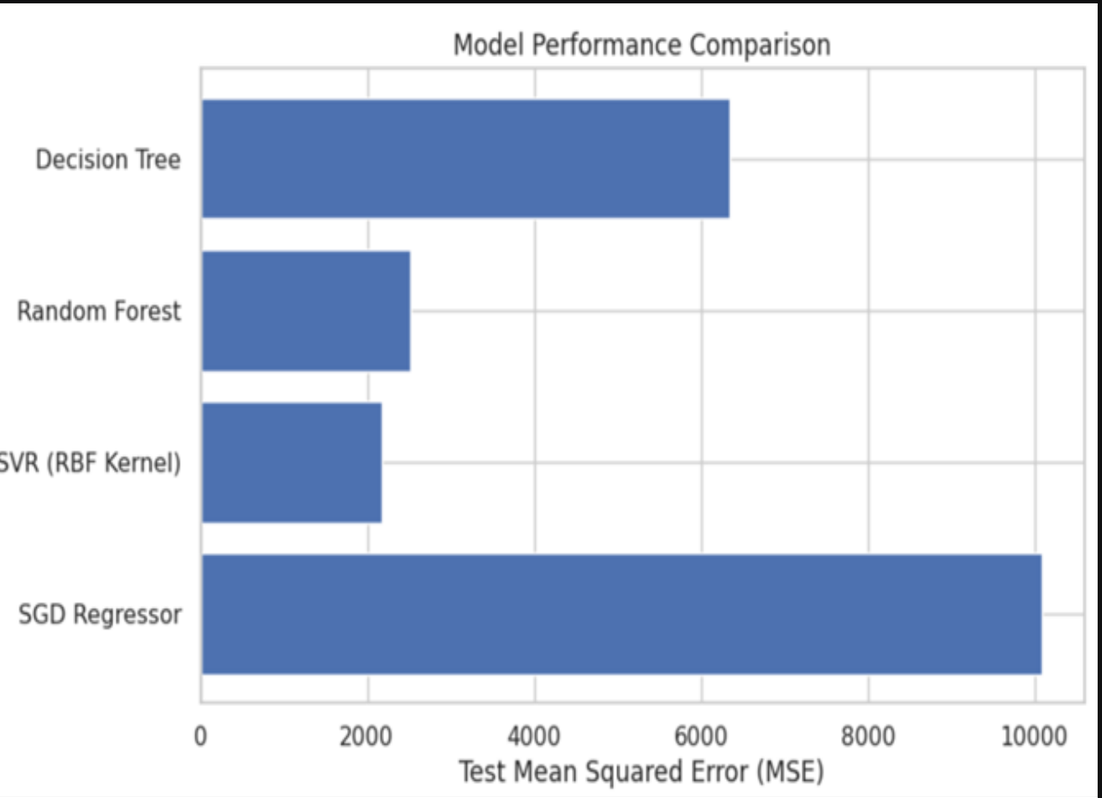


The Decision Tree shows low training error (around 2,000 MSE) on small datasets, which increases and stabilizes around 4,000 MSE as training size grows, reflecting reduced memorization. Validation error starts very high (>11,000 MSE) and decreases to 6,300–6,500 MSE with more data, indicating improved but limited generalization. The gap between training and validation errors demonstrates overfitting and high variance.



The SVR (RBF) model shows a steady decrease in training error from approximately 4,300 MSE to 1,600 MSE as the training set size increases, indicating improved model stability with more data. The validation error drops sharply from around 6,800 MSE to 2,100 MSE, demonstrating strong gains in generalization. The relatively small and narrowing gap between training and validation errors suggests low variance and good bias–variance balance. This behavior confirms that SVR with an RBF kernel effectively captures the nonlinear relationships in hourly bike demand, explaining its superior predictive performance.

	Model	Test MSE	Test MAE	Test R^2
2	SVR (RBF Kernel)	2169.112221	29.044629	0.931499
1	Random Forest	2524.563174	31.280141	0.920274
0	Decision Tree	6352.227932	53.584863	0.799395
3	SGD Regressor	10089.143616	74.001671	0.681383



Best-Performing Model Selection:

Based on the test-set performance above, we selected Support Vector Regression (SVR) with an RBF kernel as the best-performing model.

1.The SVR achieved the lowest Test MSE (2169.11) indicating the smallest overall prediction error.

2.The SVR also produced the lowest Test MAE (29.04), meaning its average hourly prediction error was approximately 29 bike rentals, which is acceptable for real-world demand forecasting.

3.SVR obtained the highest Test R^2 (0.9315), explaining more than 93% of the variability in hourly bike demand.

The SVR model demonstrated strong generalization capability, as indicated by the small difference between validation and test metrics:

Validation MSE: 2094.35 vs Test MSE: 2169.11

Validation R^2: 0.9356 vs Test R^2: 0.9315

This consistency indicates that the model is not overfitting and performs reliably on unseen data.

Best-Performing Model: SVR (RBF Kernel)

Lowest Prediction Error

Achieved lowest Test MSE (2169.11) and Test MAE (29.04).

Highest Explanatory Power

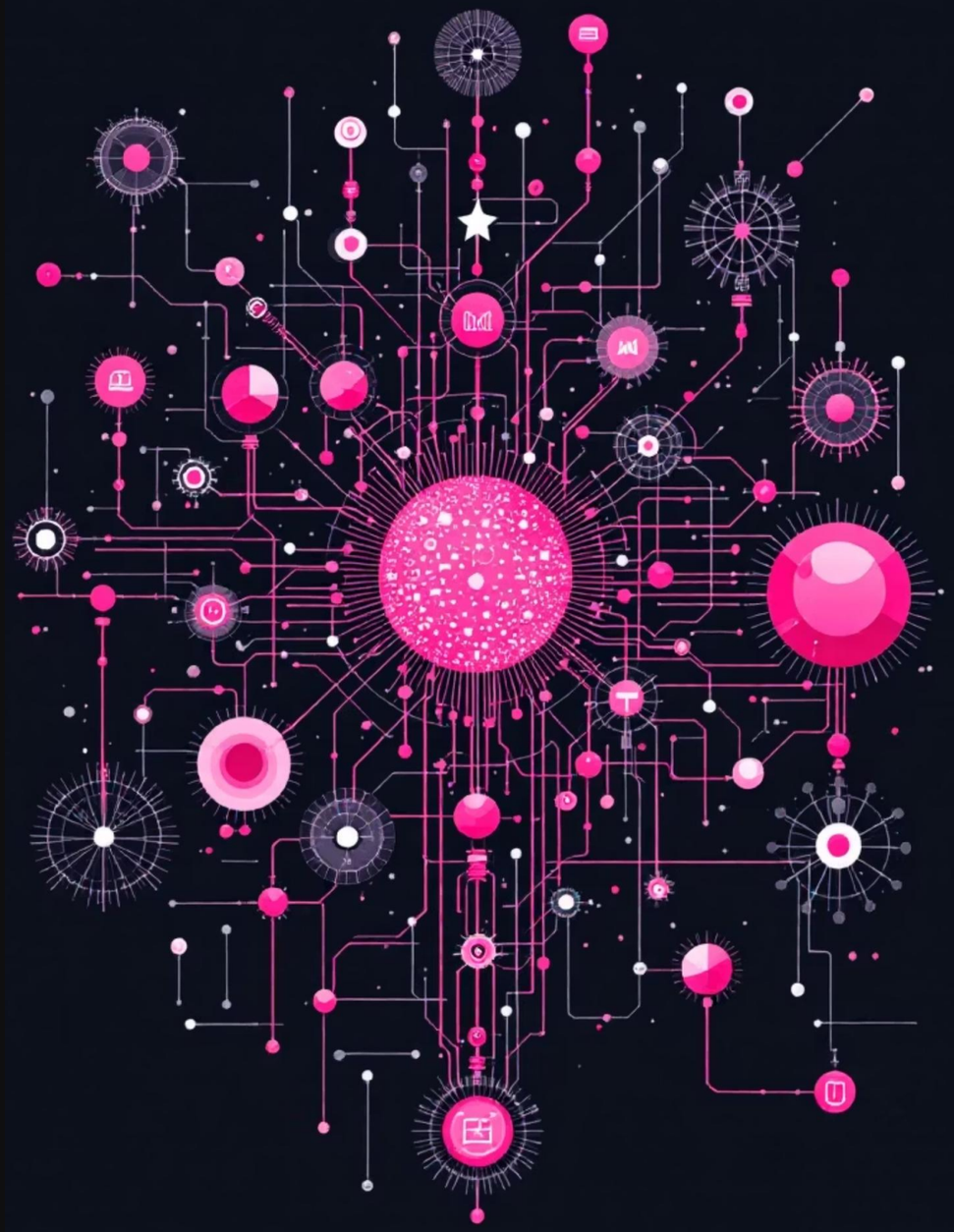
Obtained highest Test R^2 (0.9315), explaining over 93% of variability.

Strong Generalization

Consistent validation and test metrics (MSE: 2094.35 vs 2169.11; R^2: 0.9356 vs 0.9315).

Nonlinear Capability

RBF kernel effectively models complex, nonlinear demand patterns.



Explanation of Superior Performance:

Hourly bike-sharing demand is influenced by nonlinear relationships, such as the interaction between weather conditions, time of day, and type of day. The RBF kernel used in SVR is suited for modeling nonlinear patterns by projecting input data into a higher-dimensional space.

In comparison, the Decision Tree showed weaker generalization due to its sensitivity to data splits and limited ability to model smooth relationships. The SGD Regressor, being a linear model, failed to capture complex nonlinear patterns in demand. The Random Forest performed strongly by reducing variance through ensemble learning. However, the Test MSE and R^2 were still inferior to those of SVR.

Regarding Computational Efficiency and Model Complexity, SVR with an RBF kernel is computationally more expensive than simpler models such as Decision Trees or SGD. Despite this, the performance gains justify its selection.

Conclusion:

In conclusion, we selected Support Vector Regression with an RBF kernel as the most suitable machine learning algorithm for predicting hourly bike-sharing demand. It achieved the lowest prediction errors and highest explanatory power among all evaluated models while maintaining good generalization performance. These results align closely with our problem objective of accurately modeling complex, nonlinear demand patterns in an urban bike-sharing system.